

Question Number	Answer	Additional guidance	Mark
6(a)(i)	An explanation that includes the following points: • (due to) crossing over / {independent / random} assortment (1) • (crossing over) results in different allele combinations (in the chromosomes) (1) • (independent assortment) results in different combinations of {chromosomes / alleles} (in gametes) (1)	ACCEPT crossing over results in recombinant chromosomes ACCEPT alleles are exchanged reject gene ACCEPT (independent assortment results in) {different / random} arrangement of (maternal and paternal) chromosomes ACCEPT description of random assortment e.g. (homologous) chromosomes randomly aligning at equator ignore above and below equator	(3)

Question Number	Answer	Additional guidance	Mark
6(a)(ii)	An answer that includes two of the following points: • (at the beginning of) interphase the DNA is {in the nucleus / surrounded by nuclear membrane / surrounded by nuclear envelope} whereas it is not at the end of prophase (I) (1) • DNA is uncoiled at the beginning of interphase whereas it is condensed at the end of prophase (I) (1) • DNA has not been replicated at the beginning of interphase whereas it has been replicated at the end of prophase (I) (1)	ignore references to {presence/absence} of {nucleus/nuclear envelope} without linking to location of the DNA reject nucleolus ACCEPT it is {condensed / coiled / chromosomes / chromatids} at the end of prophase but not at the beginning of interphase / converse ACCEPT it is chromatin in interphase and {chromosomes / chromatids} in prophase ignore visible ACCEPT there is {double the quantity of DNA / $4n$} at the end of prophase than at the beginning of interphase / converse	(2)

Question Number	Answer	Mark
6(b)	The only correct answer is B <i>A is not correct because there would be 3 picograms in each of cell B's daughter cells</i> <i>C is not correct because there would be 6 picograms in each of cell A's daughter cells</i> <i>D is not correct because there would be 3 picograms in each of cell B's daughter cells</i>	(1)

Question Number	Answer
* 6(c)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. advantages of preventing self-fertilisation (date palm / orchid) • increased genetic variation (of seeds / new plants) / increased genetic diversity of population / reduced risk of inbreeding (depression) • benefits of increased genetic diversity e.g. idea that not all susceptible to {new selection pressure / change in the environment} / increased chance of survival for more (already) adapted individuals • chemical releasing plants can produce both ovules and pollen / chemical releasing plants prevent growth of pollen tube carrying male nuclei to ovule disadvantages of preventing self-fertilisation (date palm / orchid) • reduced chance of {pollination / fertilisation} • they need both male and female date palm plants in habitat / consideration of lack of other plants of same species nearby / low chance of wind pollination • consideration of lack of insect pollinators • if (successful) pollination doesn't occur then there will be no {seeds / new plant} production / could lead to {plant / species} becoming {endangered / extinct} / increased risk of becoming {endangered / extinct} / population could decrease • offspring may not inherit advantageous alleles / mutations could cause some pollen from another plant to not develop a pollen tube self-pollination advantages (fireweed and dandelion) • (guaranteed) production of {seeds / new plants / clones} (even if no pollination has occurred) / guaranteed (self)fertilisation • reduced risk of extinction / idea of (more) likely to {increase / maintain population} • dandelion egg cells are genetically different so resulting seeds will be genetically different self-pollination disadvantages (fireweed and dandelion) • {no / reduced} genetic variation in offspring • disadvantage of {no / reduced} genetic variation e.g. idea that all could be susceptible to a {new selection pressure / change in the environment} / genetic diseases / inbreeding depression / reduced evolution (of species)

Level 0	0	No awardable content
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.